

Giovanni Duca

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Education

University of Trento, CIMeC

Trento, Italy

M.Sc. in Cognitive Science

2023 – 2026

Computational and theoretical modelling of Language and Cognition

- Cognitive Science: Human perception and cognition
- Vision: Embodiment; representation of concepts
- Machine Learning: Main architectures and algorithms; explainability

Orientale University of Naples

Naples, Italy

B.A. in Comparative Languages and Cultures

2018 – 2022

- Discourse Analysis: Critical texts analysis (papers, movies, social media, etc.)
- Comparative Literature: Intertextual analysis and applied critical thinking of literatures
- Cultures and Society, contrastive perspectives, narrations between east and west

Experience

Inria, ALMAAnaCH

Paris, France

Supervised by Cassell, J. and Mohapatra, B.

Mar 2025 – Dec 2025

- **Thesis Research:** Machine imagery to improve agentic reasoning in situated conversational contexts.
- **Research Intern:** Creation of synthetic data to enhance conversational grounding in dialogue systems, improving interaction by dynamically adapting to user comprehension via RL.

Projects

synth4indiref 📄: Synthesize NL dialogues based on algorithmically generated complex scenarios and create challenging questions about them as a proxy for comprehension.

FedX-mod 📄: Federated Learning with SimSiam as a lightweight alternative to original contrastive method; compare performance on CIFAR-10 and SVHN.

pythia4sylo 📄: Probe an open source LLM inside a probabilistic heuristic theory of syllogistic reasoning by analysing its answers to a set of syllogisms and comparing them against human results.

Fine-tuning LLMs 📄: Fine-tuning LLMs for various NLP tasks (sentiment analysis, text classification) using standard fine-tuning, LoRA (Low-Rank Adaptation), and quantization.

Publications

MLLMs Construction Company: Investigating Multimodal LLMs' Communicative Skills in a Collaborative Building Task (aclanthology.org) 📄 Sept 2025

Marika Sarzotti*, **Giovanni Duca***, Chris Madge, Raffaella Bernardi, Massimo Poesio
(*CLiC-it 2025: Eleventh Italian Conference on Computational Linguistics*)

Certifications

CS50x - Introduction to Computer Science (HarvardX) (Certificate 📄)

Dec 2022

IELTS - C1 Academic, 8.0 Overall

Oct 2022

Apple Foundation Program

Dec 2021

Skills & Interests

Programming: Python (PyTorch + [data analysis]), Git, Slurm, HTML, CSS, SQLite

Languages: Italian (Native), English (C1), Spanish (B1), French (B1), Japanese (A2)

Volunteering: Experience as tutor and student representative